

University of Michigan

Comparative Evaluation

Micaela Ciambrone, Layilah Campbell, Seorin Jang, Dongsub Kim, Shenghao Mei

SI 622: Comparative Evaluation

March 29, 2026

Executive Summary

The comparative analysis evaluated the USDA APHIS ACIR portal against four comparators produced including eCFR, APHIS Manuals, Microsoft Learn and GOV.UK. These systems were selected because they all support users in navigating dense, high-stakes information and completing complex information retrieval tasks. To conduct the study, our team used a comparative matrix and reviewed each product across shared criteria: navigation structure, search functionality, information hierarchy, scannability, terminology clarity, accessibility support and overall interaction style. We independently explored each of the comparators, and noted the area of weakness along with the strengths then synthesized finding into a side by side matrix. The method allowed us to identify patterns across systems and evaluate how well each product supports efficient searching, comprehension, and movement across related content.

The analysis revealed four key findings, first the ACIR portal's search is too rigid and does not support flexible search strategies such as country only, commodity- only or free texting searching, Second, the portal's information hierarchy and presentation make dense regulatory content difficult to scan quickly, Third, related content is well connected, which creates friction when users need to move between requirements, manuals and taxonomy information. Fourth, the portal relies heavily on technical terminology that assumes prior knowledge, which may make the system harder to use for less experienced users.

Based on these findings, we recommend improving the portal in four key ways. First, add more flexible search modes and supportive features such as spelling corrections and short results previews. Second, improve scannability by introducing collapsible sections, archers and clearer visual hierarchy Third, strengthen cross linking between related pages so users can move easily across connected information. Finally, support comprehension by adding plain language descriptions, tooltips, or short guidance for unfamiliar terms.

Introduction

The USDA ACIR portal is a government information system designed to help users locate agricultural commodity imports requirements and interpret related regulatory guidance. It's the primary user profession who need accurate, timely information to determine whether a commodity can be imported and under what conditions. Core users tasks include searching for import requirements, interpreting regulatory terminology, and navigating across related resources such as manuals and taxonomy pages. Because the portal supports high stakes decisions, its design must help users find and understand complex information efficiently.

The goal of this comparative analysis was to evaluate how well the ACIR portal supports these tasks and to identify design patterns from other systems that could improve the user experience. Our key questions were: how effectively does the portal support users in finding and interpreting complex regulatory information? Where does it create friction in search, navigation and comprehension? What design strategies from comparable systems could strengthen findability and usability. These goals were a strong fit for the comparative analysis because the method allowed us to examine the portal alongside direct, partial, and analogous comparators with similar information challenges. They were also worthwhile goal for the client because improving search, clarity and navigation would directly increase the portals usability and better support users completing important regulatory tasks.

Methods

Our selection process for the comparative analysis began with a group brainstorming session to identify products that manage high stakes, legally binding information. We established selection criteria centered on data density and user intent parity, specifically looking for systems that require users to navigate massive databases and complex filters to determine regulatory compliance. These criteria are appropriate for the study because the ACIR portal is a functional utility tool where professional users must find and interpret dense regulatory language accurately. Following the framework for competitive user experience research described by Goodman et al. (2012, p. 79), we selected these comparators to move beyond market-based competition and instead analyze the specific interaction strategies and design patterns that support successful information retrieval in high-density data environments. By prioritizing systems with shared functional challenges, we ensure our benchmarks provide actionable insights for improving the portal's information hierarchy and search functionality.

For this comparative analysis, we evaluated the USDA APHIS ACIR Portal against four comparison products representing multiple comparator types: direct, partial, and analogous. The ACIR Portal is a government information tool that supports users in locating agricultural commodity import requirements and related regulatory guidance. As a direct comparator, we selected the Electronic Code of Federal Regulations (eCFR) because it is also a government-facing regulatory information site that requires users to navigate dense, policy-based content. As a partial comparator, we included the APHIS Manuals, since they contain closely related content and support similar user goals, but differ in structure, format, and interaction style from the ACIR Portal itself. As an analogous comparator, we selected Microsoft Learn because, although it is not a regulatory tool, it provides a strong example of how large volumes of technical information can be organized through clear hierarchy, search, and navigation patterns. We also considered [GOV.UK](https://www.gov.uk) as another analogous comparator because it demonstrates how complex public-section information can be presented in a way that is more streamlined, scannable, and action-oriented.

We analyzed these products using a comparative matrix. Each product was reviewed across a shared set of criteria including navigation structure, search functionality, information hierarchy, scannability,

terminology clarity, accessibility support, and overall interaction style. These criteria were selected because they reflect the key tasks users must complete in the ACIR Portal, including locating information efficiently, interpreting regulatory guidance, and moving across related content without losing context. They were also chosen to help us evaluate not just whether information was present, but how effectively it was structured and delivered.

To conduct the analysis, group members independently explored each comparator and documented notable strengths, weaknesses, and recurring design patterns. We then synthesized our observations into the matrix to compare products side by side. The matrix helped us identify which design decisions supported findability, comprehension, and efficient task completion, and which introduced friction. Using multiple types of comparators allowed us to assess the ACIR Portal not only against similar systems, but also against products that model stronger documentation, navigation, and information delivery patterns. This process helped us generate recommendations for improvement, particularly around search, cross-linking, and scan-friendly presentation of complex guidance.

Findings/Recommendations

Following Goodman et al. (2012, p. 88), we focused on identifying the advantages each comparator provides to users and the hindrances it places in their way. Four key areas emerged. See the full comparative matrix in Appendix A.

Finding 1: Search is rigid and does not support flexible query strategies.

The ACIR Portal requires users to search by a specific country-commodity pair. There is no way to search by country alone, enter partial terms, or browse by category. Goodman et al. (2012, p. 81) note that for search-driven tools, key features include correcting spelling errors and sorting results by relevancy. eCFR supports free-text queries with post-search filtering. Microsoft Learn offers predictive suggestions and auto-correction. GOV.UK provides a persistent search bar on every page with ranked, scannable results.

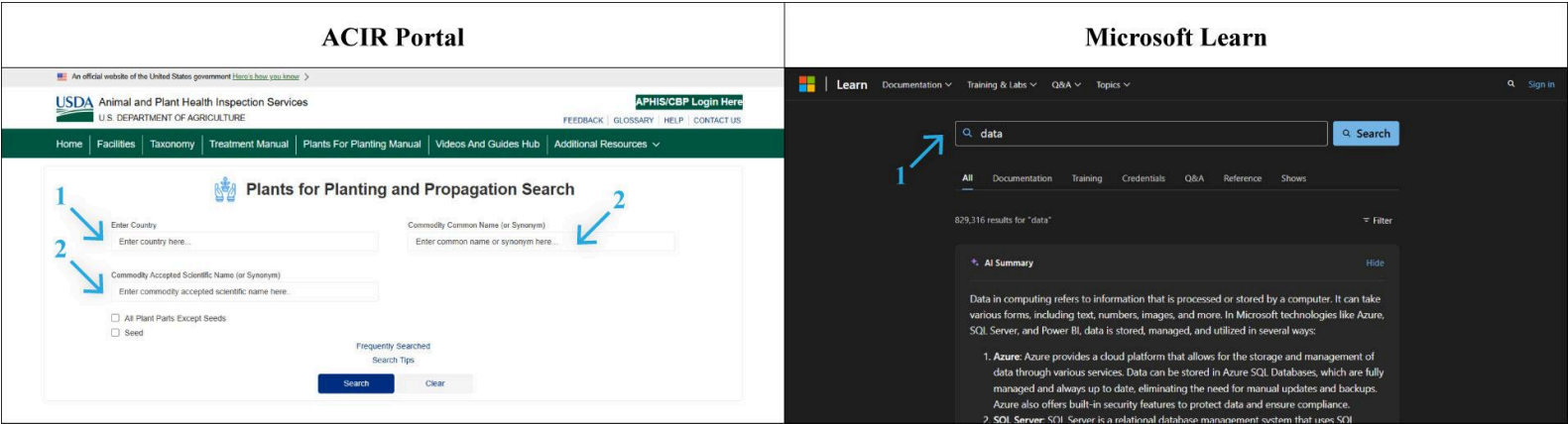


Fig 1: ACIR search interface vs. Microsoft Learn search page

Recommendation: Introduce flexible search modes supporting country-only, commodity-only, or free-text queries. Add spelling correction and short result descriptions.

Finding 2: Information hierarchy and scannability are weak.

Both the ACIR Portal and APHIS Manuals present regulatory content as dense, hyperlink-heavy text with little visual structure. Microsoft Learn handles similar volumes using collapsible sections, breadcrumbs, and in-page tables of contents. GOV.UK breaks guidance into short, task-oriented pages with bolded key terms.

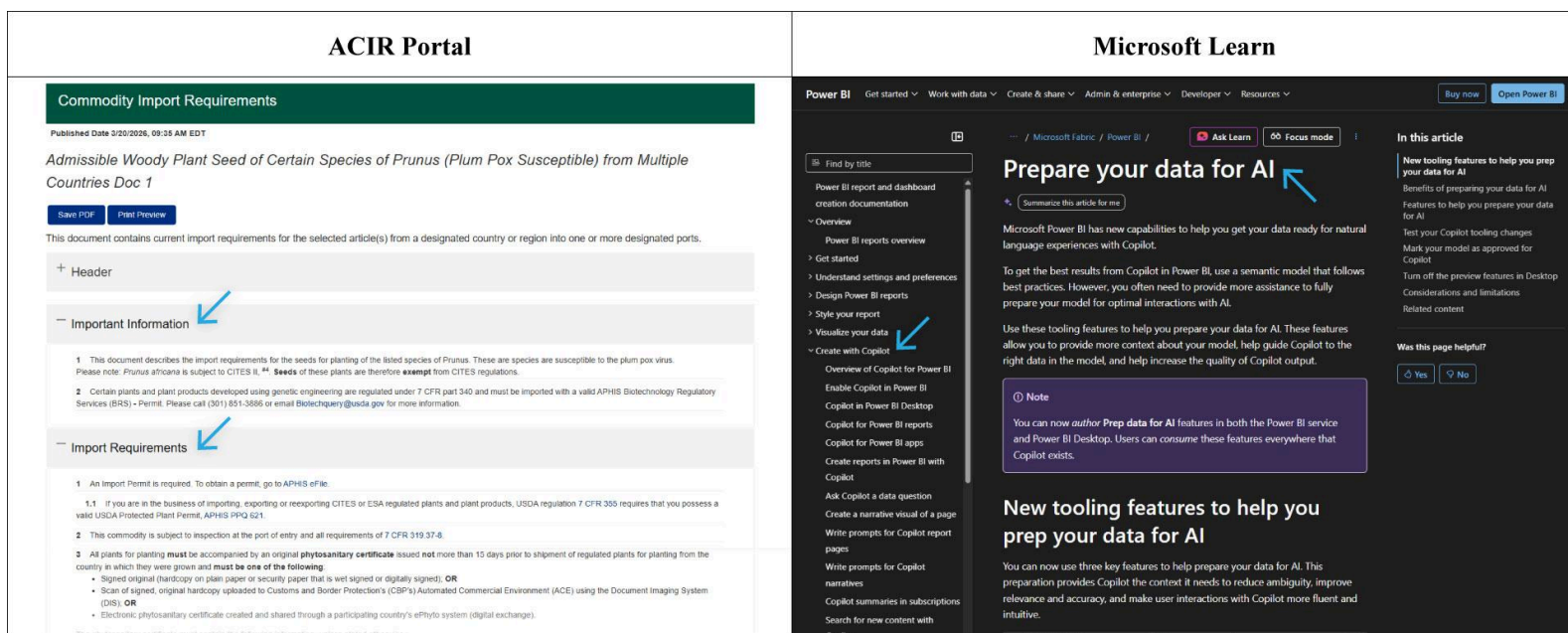


Fig 2: ACIR Portal's dense layout vs. Microsoft Learn's table of contents

Recommendation: Add collapsible sections, in-page anchors, and visual cues like bolded status indicators. Consider offering a scan-friendly layout alongside the current format.

Finding 3: Cross-linking between related content is limited.

ACIR users often need to move between search results, treatment manuals, and taxonomy pages, but few contextual links connect these areas. Microsoft Learn embeds related links and uses persistent sidebar navigation. GOV.UK includes "Related content" panels. eCFR provides inline links to referenced regulatory sections.

Fig 4: ACIR homepage tiles vs. eCFR definitions section

Recommendation: Add plain-language descriptions or tooltips alongside regulatory labels. A built-in glossary or inline definitions, similar to eCFR's approach, would bridge the gap between regulatory precision and user understanding.

Discussion

This study has several limitations. Our analysis reflects researcher judgment rather than measured user performance, as we explored comparators independently rather than testing them with actual ACIR users. The comparators also operate in different contexts: GOV.UK and Microsoft Learn serve broader audiences with larger design teams, which limits how directly their patterns can transfer to a Salesforce-based government portal. We were also unable to evaluate the authenticated CBP Resources section, so our analysis only covers the public-facing experience. Despite these constraints, the four areas identified here will directly shape our upcoming usability tests, where we plan to design task scenarios around search flexibility, information hierarchy, cross-linking, and terminology clarity to measure whether these gaps surface in real user behavior and which improvements would have the greatest impact.

References

Goodman, E., Kuniavsky, M., & Moed, A. (2012). *Observing the user experience: A practitioner's guide to user research* (2nd ed.). Morgan Kaufmann.

Appendix

Appendix A: Comparative matrix

Criteria	ACIR Portal	APHIS Manuals	eCFR	Microsoft Learn	GOV.UK
Search Flexibility	✗ Country-commodity pair only or commodity-only; no partial or free-text search	✗ No search; users must browse manually through sections or use 'ctrl+F'	✓ Free-text search across all regulations with post-search filtering	✓ Predictive suggestions, auto-correction, and related topics; AI summary of search results	✓ Persistent search bar on every page; ranked, scannable results
Spelling Correction	✗	✗	✗	✗	~

Criteria	ACIR Portal	APHIS Manuals	eCFR	Microsoft Learn	GOV.UK
	No correction; misspellings return no results	Not applicable (no search)	No built-in correction	No built-in correction	Limited; relies on keyword matching
Information Hierarchy	✗ Dense, hyperlink-heavy pages with minimal visual structure	✗ Continuous text with little heading structure or visual cues	~ Sidebar Table of Contents and section anchors, but dense text	✓ Collapsible sections, breadcrumbs, in-page Table of Contents	✓ Short task-oriented pages with bolded key terms
Scannability	~ Long pages require scrolling; Some text within collapsible sections are dense or without hierarchy	✗ No visual formatting aids; hard to scan quickly	~ In-page Table of Contents helps, but content is still dense	✓ Headings, code blocks, and callout boxes break up content	✓ Step-by-step format with clear action items per page
Cross-Linking	✗ Minimal links between search results, manuals, and taxonomy	✗ Standalone content; few links to related ACIR pages	✓ Inline links to referenced CFR sections	✓ Related articles sidebar and embedded links throughout	✓ Related content panels and breadcrumb navigation
Terminology Clarity	✗ Regulatory labels used without definitions or tooltips	✗ Expert-level language assumed throughout	~ Dedicated definitions sections per CFR part	✓ Hover tooltips and 'Before you begin' orientation sections	✓ Plain-language rewrites with inline definitions
Navigation Structure	~ Tile-based homepage; limited once inside content; no breadcrumbs	✗ Linear document structure; no sidebar or breadcrumbs	✓ Persistent sidebar Table of Contents; title/part/section hierarchy	✓ Sidebar nav, breadcrumbs, and version selectors	✓ Breadcrumbs and step-by-step navigation on every page

✓ = Supported ✗ = Not supported ~ = Partial support